Chinmay Arvind

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Education

University of Minnesota - Twin Cities Expected May 2027
Master of Science in Computer Science
Minneapolis, MN

The University of British Columbia May 2025
Bachelor of Science in Computer Science (Honors), GPA: 3.85
BC, Canada

Publications & research

DynamicMem: A Long-Horizon Memory Benchmark with Evolving User Context Jan – Sep 2026
NeurIPS · under review

- Co-authored an LLM evaluation benchmark for long-horizon memory, designing evaluation methodology, analyzing model failure modes, and synthesizing findings to improve reliability in conversational foundation-model systems.

Undergraduate Honors Thesis Sep 2024 – Apr 2025
PNRL: Reinforcement Learning Development for an Undergraduate AI Course

- Engineered a modular client-server Java-based RL system capable of processing 10,000+ training episodes per session with efficient Q-table updates, enabling students to conduct real-time experimentation of RL strategies.
- Implemented an epsilon-greedy Q-learning policy, increasing final cumulative rewards of the RL agent by 15% within the first 100 training episodes.
- Built a Python-Java bridge via Py4J, enabling 15+ teams to use Python ML libraries in a Java game—cutting development time by 50%.

Achievements

Carlson School of Management — Analytics for Good Hackathon Jan 2026
2nd Place

- Placed 2nd by delivering an end-to-end analytics prototype and communicating nuanced ideas clearly to non-technical stakeholders, translating technical findings into concise, actionable recommendations.

Relevant coursework

The University of British Columbia
Algorithms · Data Structures · Software Engineering · Databases · Parallel Computing · Machine Architecture · Machine Learning · Deep Learning · Artificial Intelligence · Linear Algebra · Probability · Regression Analysis

Experience

Vidasmi

Jun 2026 – Sep 2026

AI Engineer Intern

Remote

- Built an autonomous drone mission platform combining natural-language planning, real terrain/weather data, PX4 flight execution, and GPT-5 Mini advisory, achieving 100% safe recovery across 200+ fault-injected physical flights.
- Designed a governed multi-agent LLM and deterministic planning architecture integrating intent extraction, allowlisted strategy generation, route construction, terrain-aware altitude planning, geofencing, real-weather ingestion, aircraft feasibility, energy-reserve analysis, route scoring, bounded refinement, human approval, and preflight authorization, achieving 98% end-to-end workflow success and 96% validation coverage across registered mission constraints while automatically rejecting unsafe missions.
- Reduced drone mission-planning time by 80% (10→2 minutes) by building an LLM-powered platform that converts inspection requests into multi-step missions with automated route planning and terrain, weather, and battery validation.
- Engineered a production-style mission-execution backend with durable workflows and single-use, revision-bound authorization to block duplicate or unapproved launches, preserve mission state across workflow retries, and retain request-to-flight audit trails for failure investigation and recovery.
- Cut repetitive flights by 60% by implementing LiDAR-based obstacle detection, wind-aware route adaptation, and operator-approved replanning that enables interrupted missions to continue.
- Scaled the control plane for 1,000+ mission requests per day and 50+ concurrent planning workflows, sustaining sub-5-second median mission-generation latency and sub-15-second p95 end-to-end planning latency while executing terrain, geofence, weather, aircraft, and energy validation.

University of Minnesota

Sep 2025 – May 2026

Graduate Research Assistant

Minneapolis, MN

- Built a multimodal financial-fraud research system processing 10+ GB production datasets and evaluating models over 3,200 invoice and payment cases; combined document vision, financial text, transaction features and heterogeneous graphs; achieved 0.871 AUROC, 0.81 AUPRC, and 84% Recall@Top-5%, improving AUPRC by 19% over the strongest unimodal baseline with engineered logistic scoring, supported by three-seed temporal-holdout evaluation, modality ablations and paired bootstrap testing.
- Developed a bounded investigation and robust case-prioritization workflow, comparing parallel document, graph and transaction evidence gathering against a single-worker baseline, with schema-validated read-only tools, checkpointed trajectories, action budgets and deterministic guardrails; achieved 100% evidence-attribution accuracy across 300 checks, 99.6% valid tool calls, rejected 100% of unauthorized tool requests, and captured 13.16% more expected fraud loss versus probability ranking under a fixed analyst-review budget while reducing median investigation time from 14.2 to 6.0 minutes.
- Automated point-in-time feature generation, graph construction, training and evaluation through a reproducible MLOps pipeline scheduling batch workloads on Slurm-managed HPC infrastructure with versioned artifacts, checkpointed retries and model lineage, reducing retraining time from 64 to 21 minutes; implemented drift detection and statistical canary promotion gates, detecting 93% of seeded data/concept-drift events with a 4.2% false-alert rate, configuring daily scheduled runs with checkpointed retries, sustaining 99.4% successful scheduled runs, and rolling back statistically regressing candidates in approximately 0.12 seconds.

University of British Columbia May 2024 – Apr 2025

ML & Backend Engineer

BC, Canada

- Built an AI-powered educational content platform that converts PDFs and DOCX files into context-aware questions and document-grounded answers, processing 15+ GB of educational content and generating 1,000+ questions/hour; maintained 98% extraction accuracy while reducing manual question creation by 80%.
- Improved the platform's retrieval and generation pipeline through model fine-tuning, document grounding, and asynchronous processing, increasing response-generation speed by 25%, model accuracy by 15%, and question relevance by 30%; productionized the system for 100+ users at 99.1% uptime, reducing API response time by 30% and improving file-retrieval efficiency by 50%.
- Built the end-to-end ML production lifecycle spanning data ingestion and cleaning, embedding generation, model tuning and evaluation, retraining, deployment, fallback, and monitoring; added secure access controls and persistent document/question storage while reducing serving and infrastructure cost by 5% and supporting reliable production inference.

University of British Columbia Sep 2024 – Apr 2025

Teaching Assistant

BC, Canada

- Capstone Course: Mentored 120+ students across 20+ teams, advising on GitHub workflows, coding standards, and leading team consultations on architecture and feature planning—strengthening collaboration and project quality.
- Introductory Programming Course: Led labs for 30+ students; delivered personalized feedback improving assignment completion rates and student fluency in Java.

Projects

CreditLens Sep 2026

Python · FastAPI · LlamaIndex · Weaviate · Snowflake Cortex · DeepEval

- Built a permission-aware lending RAG system across 3,800+ pages of synthetic policy and borrower documents using hybrid retrieval, semantic chunking, selective query rewriting, reranking, and page-level citation enforcement, achieving 86.54% Recall@10 and 0.788 nDCG@10, a 12.49 percentage-point improvement in retrieval recall.
- Created a 240-question adversarial evaluation set spanning policy interpretation, borrower evidence, missing documents, contradictions, restricted data, and underwriting exceptions, achieving a 92.5% grounded-answer pass rate while reducing unsupported claims to 1.7%.
- Implemented response caching, policy-aware access controls, end-to-end retrieval tracing, and CI quality, latency, and cost gates, improving answer quality by 13.4 percentage points, measuring 300 ms p95 latency at 8 concurrent clients across two local API processes with warm caching, and reducing average cost per request by 31% to $0.043.

FinServe Sep 2026

PyTorch · Ray Serve · vLLM · Redis · MLflow · Airflow

- Built a distributed text and multimodal inference platform with GPU-aware routing, continuous batching, prefix caching, and speculative-decoding evaluation, increasing sustained throughput from 13.0 to 34.6 requests/s, token throughput by 2.64×, and average GPU utilization from 29% to 58%.
- Benchmarked 6,000+ inference requests across baseline and optimized serving configurations; in a separate quantized 3B release benchmark, reduced median client TTFT from 247 ms to 116 ms and p95 latency from 2.24 s to 0.86 s, achieving 100% correctness on a 32-case release regression suite.
- Automated model evaluation, deployment, and rollback orchestration with CI performance gates, an 18-task Airflow pipeline, and MLflow evidence tracking, completing 512/512 release benchmark requests and validating 60 post-deployment probes and 1→2→1 GPU engine scaling.

PayOps

Sep 2026

Python · LangGraph · Kubernetes · Prometheus · Elasticsearch · OpenTelemetry

- Built an agentic incident-response system correlating Kubernetes health, logs, traces, deployment changes, and payment telemetry across 24 reproducible failure scenarios, achieving 91.7% root-cause Recall@1 and Recall@3 in predicate-assisted development replay.
- Developed a failure-oriented evaluation harness covering OOMKills, bad rollouts, scheduler saturation, dependency outages, misleading telemetry, and payment-slice degradations, rejecting 100% of 120 unauthorized remediation attempts in component evaluations.
- Added structured investigation traces, incident memory, deterministic approval gates, GraphQL, Slack notifications, and durable Pub/Sub processing, achieving 8.41-second median diagnostic replay latency, reducing median investigation time from 11.8 min to 2.9 min, and maintaining $0.07 average LLM cost per incident.

HookGuard

Go · Rust · Redis Streams · Kubernetes · Prometheus · scikit-learn

- Sustained 12,249 payment events/sec at 46.46 ms p99 acknowledgement latency in a 60-second local benchmark using atomic deduplication, enqueueing, and asynchronous processing; all 735,006 requests were accepted.
- Produced zero duplicate risk-decision records across 1M+ replayed webhook deliveries using shared event IDs and atomic decision publication, with concurrent-retry and worker-restart verification.
- Achieved 0.95 F1 versus a 0.87 rules baseline on 2,000 held-out transactions, with 196 ns p99 in-process scoring, using a calibrated lightweight classifier and reproducible evaluation pipeline.

Risk-Adjusted Hallucination Detection

Feb – Apr 2026

Python · Qwen · Calibration · Self-consistency · Abstention

- Built a calibrated hallucination-risk detector using 4 uncertainty/evidence signals: token uncertainty, self-consistency disagreement, semantic entropy, and groundedness consistency.
- Engineered an end-to-end ML pipeline for answer generation, feature extraction, feature-table construction, detector training, calibration, abstention thresholding, frozen model bundles, and transfer evaluation.
- Evaluated on 5.3K grounded QA examples across 4K PHANTOM and 1.3K WikiQA samples using Qwen3-8B generation, Qwen3-14B judging, and k=5 self-consistency sampling.
- Achieved 77.3% AUROC, 64.5% AUPRC, 75.9% accuracy, and 63.7% F1 on PHANTOM hallucination-risk detection.
- Improved PHANTOM selective accuracy from 68.3% to 78.6% by abstaining on the riskiest 18.2% of answers while preserving 81.9% coverage.
- Compared Platt scaling, isotonic regression, temperature scaling, and raw probabilities, reaching calibrated test ECE of 0.0734 on PHANTOM and 0.0725 on WikiQA.
- Validated robustness limits through bidirectional transfer experiments, PHANTOM→WikiQA and WikiQA→PHANTOM, diagnosing feature-distribution shift, calibration instability, and threshold non-portability.

FactorForge

Sep 2026

Python · LangGraph · Polars · MLflow · Neo4j · PyTorch

- Built an autonomous quantitative-research system that turned natural-language investment ideas into relevant literature, structured hypotheses, hybrid factor strategies, sandboxed backtests, and iterative research decisions, backed by a deterministic execution engine exercised across 45 historical signal-and-cost cases covering 30,000+ historical data records.
- Developed a validation pipeline with multi-paper synthesis, walk-forward and purged validation splits, transaction-cost modeling, and statistical robustness checks to evaluate published-factor reproduction and factor-spec extraction accuracy across 15 price-based signals across 3 cost settings achieving 100% agreement across 40+ portfolio valuations and 15+ trade fills—including fees against independent LEAN execution of a reference backtest.
- Recorded successful and failed experiments as reusable research memory and training trajectories, preserving 100% of prompts, datasets, code, backtests, and experiment lineage, verifying linked research artifacts, tracing 1,200+ agent/tool spans, and reducing average research runtime from 46 min to 18 min at $0.84 average LLM cost per research run.

Jersey Number Recognition Boosted

Mar – May 2025

PyTorch · OpenCV · FlashAttention · ViTPose · Temporal smoothing

- Built an end-to-end computer vision pipeline to extract jersey numbers from sports video, combining person tracking, pose-guided ROI cropping, and OCR/scene-text recognition to produce per-player number predictions.
- Implemented pose-detection pipeline improvements by adding temporal smoothing for keypoints and enabled GPU reduced-precision inference to improve stability and performance.
- Hardened pipeline execution by improving device selection (CPU/GPU), fixing keypoint serialization for JSON outputs, tuning pose-based cropping thresholds, for reduced latency and reproducible runs.

EEG Sleep Deprivation Classification

PyTorch · EEG · CNNs · ViTs · GRU/LSTM

- Built an end-to-end EEG deep learning framework on 8,300 resting-state EEG epochs to classify normal sleep vs. sleep deprivation and predict session-level SSS/KSS sleepiness scores.
- Designed leakage-safe 70/15/15 subject-wise splits with 5,738 train, 1,246 validation, and 1,316 test samples, ensuring no subject overlap across train, validation, and test sets.
- Demonstrated transfer of learned EEG representations to downstream sleepiness prediction, achieving 0.692 Pearson and 0.738 Spearman correlation on held-out subjects using feature-based KSS modeling with leakage-safe subject-wise splits.
- Benchmarked 4 EEG representation learners—Residual EEG CNN, 2-Branch EEG CNN, DeiT-Tiny, and ViT-Small—and reused the best encoder for downstream sleepiness modeling.
- Implemented 5-fold subject-wise cross-validation to evaluate generalization across held-out subjects and prevent identity leakage in EEG experiments.
- Built downstream prediction pipelines using pretrained EEG embeddings with Ridge/MLP/GRU/LSTM models, direct ordinal prediction, and multi-task learning for joint classification and sleepiness estimation.
- Improved training robustness with AdamW, ReduceLROnPlateau, early stopping, dropout, weight decay, label smoothing, class-weighted loss, weighted sampling, time jitter, and random time-crop augmentation.

Drift.ai Apr 2026

Next.js · FastAPI · Supabase · Ollama · Qwen

- Built Drift.ai, a full-stack AI personal finance platform that replaced a fragmented 7-step manual analysis workflow with a unified product flow across 2 transaction ingestion paths (CSV + Plaid), automated drift detection, what-if simulation, and PDF/email report delivery.
- Engineered a hybrid deterministic + LLM pipeline that combined BLS CPI-U inflation-adjusted spending analysis with local Qwen via Ollama to run multi-turn behavioral interviews, classify spending into 6 behavior tags, and generate personalized 30-day recovery plans backed by 8 cited financial evidence fields.
- Implemented a production-style, privacy-first architecture with Auth0, Stripe Checkout, Resend, Vercel, and a 4-service Dockerized stack; kept raw transaction rows local while persisting only 6 summary data entities in Supabase, enabling monetized report delivery with minimal sensitive-data retention.

CleanBreak

TypeScript · Next.js · Playwright · SQLite · Solari Desktop

- Built an autonomous subscription-cancellation platform that reduced average cancellation time from 8 minutes to 45 seconds, a 91% reduction, automating provider navigation, retention-offer handling, guarded execution, and independent billing verification.
- Helped 10 pilot users avoid $2,400 in annual subscription renewals by converting unwanted recurring expenses into independently verified cancellation outcomes with durable receipts.
- Reduced cancellation-related manual follow-up by 80% through automated billing-state verification and auditable receipts confirming that future renewal charges had stopped.

Disagreement-Aware RAG Aug 2025

FastAPI · LlamaIndex · Next.js · Hugging Face Transformers

- Built a disagreement-aware RAG system that abstains when predicted disagreement risk is high.
- Trained a lightweight logistic regression model over self-consistency variance (k-resampled generations), evidence overlap (ROUGE-L–style), and an token-level entropy
- Engineered a full-stack app that allows querying of questions on a test dataset and plots coverage vs. hallucination; achieved ~97% coverage @ 3–4% hallucination ($\tau \approx 0.55$–0.60).
- Implemented a zero-shot entailment auditor with BART-MNLI to flag likely hallucinations in evaluation.

AIvaluate Jun – Aug 2024

JavaScript · Node.js · Express.js · React.js · Docker

- Led a team in developing a course platform integrating OpenAI's Assistants API to automate grading and feedback for student submissions—reducing grading time by 70% and increasing grading consistency.
- Implemented a microservices architecture with robust user authentication and role-based access control to ensure modularity, fault isolation, and streamlined continuous integration and deployment.
- Leveraged test-driven development (TDD) across services, increasing test coverage to 90% and reducing post-deployment bugs by 40%.

Algorithmic Trading System Jan – Feb 2025

FastAPI · PyTorch · scikit-learn · Pandas · NumPy

- Developed a stock trading simulation platform combining market trends, news sentiment, and ML models to mimic real-world trading—enabling informed, data-driven strategies.
- Constructed a hybrid model using K-Means, Agglomerative Clustering, and logistic regression to classify market regimes—improving detection accuracy by 34%, enabling faster strategy shifts and boosting simulated returns.
- Designed a CNN-LSTM model with reinforcement-based position sizing, increasing prediction accuracy by 12% and boosting simulated returns by 30% during backtesting.
- Built a sentiment-analysis pipeline from RSS feeds to generate real-time trading signals and enhanced predictive insight.

Technical skills

Languages: Python, TypeScript, Java, C, R, JavaScript, SQL, HTML, CSS

LLMs, agents & retrieval: LangGraph, LangChain, LlamaIndex, Hugging Face Transformers, Sentence Transformers, Ollama, OpenAI API, Snowflake Cortex Search, NLTK, OpenSearch, Elasticsearch, FAISS, Weaviate, BM25, Hybrid Retrieval, Reciprocal Rank Fusion, Semantic Chunking, Query Grounding, Cross-Encoder Reranking

Machine learning: PyTorch, scikit-learn, XGBoost, NumPy, Pandas, SciPy, Polars, OpenCV, CUDA, Matplotlib, Seaborn, PyTorch Geometric, BERT, Vision Transformers (ViT), GraphSAGE, Multimodal Learning, Late Fusion, Probability Calibration

Backend & APIs: FastAPI, Flask, Node.js, Express.js, Uvicorn, Pydantic, SQLAlchemy, HTTPX, REST APIs, GraphQL, gRPC, Protocol Buffers

MLOps & observability: Apache Airflow, MLflow, Artifact Versioning, Checkpointed Pipelines, Statistical Canary Gates, Drift Detection, DeepEval, RAGAS, OpenTelemetry, Prometheus, Grafana, LangSmith, Temporal Validation, Modality Ablation, Paired Bootstrap Testing

Databases & infrastructure: PostgreSQL, MySQL, SQLite, Redis, Supabase, Docker, Docker Compose, Kubernetes, Terraform, AWS ECS/Fargate, ECR, ALB, CloudFront, CloudWatch, IAM, SQS-Compatible Messaging, Hugging Face Spaces, Hugging Face Hub, Render, AWS CloudFormation, Slurm, High-Performance Computing (HPC)

Data, quantitative systems & tooling: LEAN, PySpark, Apache Spark, PyArrow, Parquet, ReportLab, Pillow, Bun, D3.js, pytest, pytest-cov, Ruff, mypy, Git, GitHub Actions, DroneCI, uv, JUnit